Fabian Lander

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Education

PhD Mathematics *2024–May 2027 (expected)*
Max Planck Institute for Mathematics in the Sciences (International Max Planck Research School), degree from Leipzig University
Research Focus: Vertex-minimal paper tori (polyhedral embeddings of flat tori)
Advisor: James Farre

MSc Mathematics *2021–2023*
Heidelberg University, final grade 1.3 (German scale: 1.0 best, 4.0 lowest pass)
Thesis: Vertical Foliations of Triangulable Dilation Tori
Advisor: James Farre

BSc Mathematics *2017–2021*
Heidelberg University, final grade 1.8
Thesis: Polygonal Symplectic Billiards
Advisor: Peter Albers

Work Experience

Research Assistant (Mathematics Lab, Leipzig) *2024–Present*
Paid position alongside the PhD
Developing interactive visualizations for mathematical research and outreach
Collaborating with researchers on visual tools for geometry and dynamics
Supporting public engagement through exhibits and demonstrations

Research Assistant (Heidelberg Experimental Geometry Lab) *2022–2024*
Developed real-time visualization tools for flat surfaces using WebGL/Three.js

Research Assistant (Collaborative Research Center SFB 191, Heidelberg) *2021–2024*
Developed high-performance symplectic billiard dynamics simulator (C++/OpenGL)
Co-authored a paper on periodic orbits in symplectic billiards
Implemented discrete dynamical system visualizations

Teaching

Teaching Assistant in Analysis 1–3 (Heidelberg University) *2020–2022*
Led my own tutorial sections of 20–30 students (two to four sessions per week) in Real Analysis and Measure Theory, for lecture courses of about 300 students
Planned and taught every session; graded the weekly homework and the final exams

Undergraduate Project Supervision (Heidelberg Experimental Geometry Lab) *2022–2024*
Supervised groups of one to five undergraduates on projects in geometric computing

Outreach and Service

Organizer, Reading Seminar on Riemann Surfaces (MPI MiS, Leipzig) *2025*
Fifteen weeks on Gunning's *Lectures on Riemann Surfaces* with three PhD students and three postdocs; led five of the sessions

Participant, Trimester at the Institut Henri Poincaré, Paris *Spring 2026*
"Illustration as a Mathematical Research Technique"; co-wrote a report on it for the *Mitteilungen der DMV* (see Publications)

Long Night of the Sciences, Leipzig *2025*

Showed 3D-printed models from my research at the stand of the Mathematics Lab

Outreach Events, Student Seminar of the Heidelberg Experimental Geometry Lab *2022–2024*
Helped run outreach events and showed students how to 3D print and visualize mathematical objects

Publications and Preprints

A Universal 8-Vertex Origami Torus Triangulation, with Peter Doyle, Richard Evan Schwartz and Steve Trettel, preliminary version (2026), fabianlander.github.io/papers/universal-8-vertex.pdf.

The two most symmetric flat tori as eight-vertex paper tori, arXiv:2609.10703 (2026). 26 pages. Preprint.

Symplectic Tiling Billiards on Complete Affine Tori, with Charles Daly, arXiv:2608.28894 (2026). 47 pages. Preprint.

Illustrating Hyperbolic Surfaces with Mesh Embeddings, with Erik Löffelholz, Diaaeldin Taha, Steve Trettel and Anna Wienhard, in *Proceedings of Bridges 2026: Mathematics, Art, Music, Architecture, Culture*, pp. 197–204, Bridges Organization, 2026.

Immersive Visualization of Flat Surfaces Using Ray Marching, with Diaaeldin Taha, in *Proceedings of Bridges 2025: Mathematics and the Arts*, pp. 191–198, Bridges Organization, 2025.

Symplectic billiards for pairs of polygons, with Peter Albers and Jannik M. Westermann, arXiv:2402.12244 (2024). 38 pages. Accepted for publication in the *Arnold Mathematical Journal*.

Illustration als mathematische Forschungstechnik, with Tim Reinhardt and Martin Skrodzki, *Mitteilungen der Deutschen Mathematiker-Vereinigung* 34 (2026), no. 3, 153–158.

In Preparation

Seven-vertex paper tori in $\mathbb{R}^4$ (working title), with Steve Trettel.

Surfaces of revolution in homogeneous three-manifolds (working title), with Steve Trettel.

Selected Software Projects

Eight-Vertex Paper Tori Verification *Python, 2026*
Exact-arithmetic verification of every claim in the eight-vertex paper tori preprint, over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{3})$
github.com/FabianLander/eight-vertex-paper-tori-verification

Interactive Pseudosphere with Hyperbolic Tiling *Three.js/GLSL, 2026*
Real-time (2,3,7) triangle tiling on surfaces of revolution with adjustable ambient curvature, using RK4 ODE integration
fabianlander.github.io/talks/rotationally-symmetric-surfaces

Flat Tori Gallery *Three.js/WebGL, 2026*
Interactive isometric embedding experiments of flat tori in $\mathbb{R}^3$ and S^3
fabianlander.github.io/demos

Symplectic Billiards Simulator *C++/OpenGL, 2020*
High-performance research tool for dynamical systems analysis
github.com/SiegfriedFabian/Billiards